

Astronomical Object Classification Using Machine Learning on SDSS17 Data

1. Contribution & Roles table

Contributor	Student ID	Student_mail_id	Role
Aadesh Sunil Kshetre	A00052071	aadeshsunil.kshetre2@mail.dcu.ie	Coding(Merging of code),reviewed the document and added the points,contributed to the design and development of (PPT)
Shubin Li	A00048674	shubin.li2@mail.dcu.ie	Coding (supplementary implementation and extended analysis); Report (contributed to the Error Analysis and Limitations sections).
Yu Yu Lwin	A00053026	yuyu.lwin2@mail.dcu.ie	Entire report creation by analysing on the coding and results
Varad Abhay Mahadik	A00051764	varadabhay.mahadik2@mail.dcu.ie	Coding(Wrote the core codebase for data processing, modelling.); Designed and prepared the entire project PPT

2. Title & Abstract

2.1. Title

Astronomical Object Classification Using Machine Learning on SDSS17 Data

2.2. Abstract

This project is mainly focused on “Astronomical Object Classification Using Machine Learning on SDSS17 Data” explored Stellar Classification using statistical and machine learning to identify and categorize celestial objects such as the stars, galaxies, and quasars [2], depending on their characteristics. In astronomy, the light emitted by each star or galaxy varies in intensity and wavelengths. By examining these characteristics of light, what type of object can be determined.

This investigation highlights features, models and training, error analysis, limitation and further improvement.

3. Problem & Data

3.1. Problem Definition

The main goal of this project is to build a system that can automatically classify these objects (stars, galaxies, and quasars) into their correct categories using data collected by telescopes by using the dataset from Stellar classification dataset - SDSS17 [2] dataset. Model development and comparison between them, evaluation metrics for accuracy, F1 score, recall, precision, and classification report are included in this problem solution report.

3.2. Task and Metrics

The main task is to develop and train a machine learning model to classify each object into one of the three categories using statistical and machine learning techniques to train the model. Performance is measured using the metrics such as accuracy, precision, recall, F1-Score.

3.3. Dataset description

The dataset included 100,000 observations of space objects used from the SDSS. There are 18 columns included in total such as obj_ID, alpha (Right Ascension angle (at J2000 epoch)), delta (Declination angle (at J2000 epoch)), u (Ultraviolet filter in the photometric system), g (Green filter in the photometric system), r (Red filter in the photometric system), i (Near Infrared filter in the photometric system), z (Infrared filter in the photometric system), redshift (redshift value based on the increase in wavelength) and so on.

Every observation is described by 17 feature columns, numerical or categorical variables such as obj_ID, alpha, delta, spec_obj_ID and so on and 1 class column (class), target variables, which identifies it to be either a star, galaxy, or quasar.

3.4. Source Code Link

[Astronomical Object Classification\(Codebase\).ipynb](#)

[Astronomical Object Classification\(Supplementary & Extended Analysis\).ipynb](#)

Dataset Link

<https://www.kaggle.com/datasets/fedesoriano/stellar-classification-dataset-sdss17>

4. Methods

4.1. Data Preprocessing

Data cleaning and preparation are needed before training the model. Firstly, check the dataset for missing values or irrelevant columns first.

Then checking the unique identifier columns such as ['obj_ID', 'run_ID', 'rerun_ID', 'field_ID', 'spec_obj_ID', 'fiber_ID'] and dropped since they will not include to the performance of the model. The reason is that those unique identifiers could lead to overfitting the model. We can consider the data analysing using those identifier columns as inputs in a simple model training as an alternative approach.

After that, using IQR (Interquartile Range) method to eliminate outliers from the feature columns and to improve the model stability.

Finally, MinMax Scaling is applied to normalize numerical features in the dataset. This normalization process recalls the data for each numerical feature to transform a range between 0 and 1. This is to ensure that all features have the same scale and normalizing helps to improve the machine learning model performance.

4.2. Models

In this project, various models are considered to evaluate the metrics such as Logistic Regression as a baseline, Decision Tree Classifier, Random Forest Classifier, K-Nearest Neighbors (kNN), Support Vector Machine (SVM), Gradient Boosting Classifier, Adaptive Boosting (AdaBoost), XGBoost for comparison. Among all models in our project, XGBoost provides the best performance with highest accuracy, precision, recall, and F1-score.

4.3. Validation Protocol

A two-step validation strategy was used in this project. First, the dataset was split into a training set (70%) and a test set (30%) using the ‘train_test_split’ function with a fixed random state to ensure reproducibility. The training set was evaluated using 5-fold cross validation. Specifically, the KFold object was initialized with 5 splits, shuffled and with a fixed random state, then the average accuracy of the cross validation was calculated using the ‘cross_val_score’ function. This provides a robust estimate of the generalizability of each model to data not seen during training.

After cross validation, each model was retrained on the entire training set and evaluated once on the unseen test set. The following performance metrics were calculated for this test set: accuracy, weighted F1 score and confusion matrix. This protocol ensures that model selection is based on cross validation performance, and that the final results are reported based on a completely separate test set that was not included in training or cross validation.

4.4. Supplementary Implementation and Extended Analysis

In addition, we performed supplementary implementation and extended analysis to further explore model behavior and performance. Key aspects include:

- Outlier handling: comparison of model performance under three different scenarios.
- Feature selection: mitigation of multicollinearity and exploration of different feature sets.
- Visual comparison of accuracy across each feature combination.
- Cross-validation to assess model stability.
- Includes error analysis, limitations, and insights for future work.

5. Results (metrics, plots, comparisons vs. baseline)

5.1. Evaluation MetricsTable comparison between Codebase and Supplementary Analysis

5.1.1.Codebase Analysis

	Models	Accuracy	Precision	Recall	F1_score
0	LogisticRegression	0.910640	0.910775	0.910640	0.907931
1	Desicion Tree	0.958675	0.958911	0.958675	0.958788
2	Random Forest	0.958675	0.958911	0.958675	0.958788
3	K Nearest Neighbours	0.897792	0.898565	0.897792	0.894861
4	Support Vector Machines	0.931347	0.931123	0.931347	0.929724
5	Gradient Boosting	0.973731	0.973223	0.973731	0.973221
6	AdaBoost	0.915320	0.921388	0.915320	0.917620
7	XGBoost	0.973289	0.972889	0.973289	0.972987

Image 1: Model Comparison for DataFrame (Codebase Analysis)

5.1.2.Supplementary and Extended Analysis

Following supplementary analysis, the accuracy of three models improved: logistic regression, decision trees, and random forests increased by 1.0%, 0.8%, and 2.1%, respectively.

	Model	Before	After	Point Increase	Percent Increase
0	Logistic Regression	0.910640	0.919673	0.9	1.0%
1	Decision Tree	0.958675	0.966095	0.7	0.8%
2	Random Forest	0.958675	0.978646	2.0	2.1%

Image 2: Model Comparison based on Supplementary Analysis

5.2. Performance Comparison for Models vs. baseline

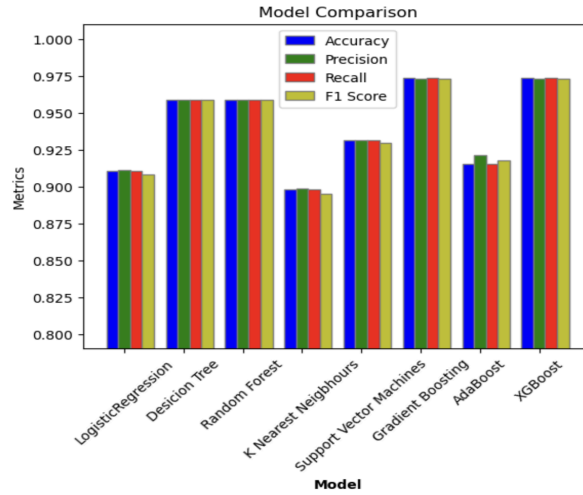


Image 3: Bar plot for Model Comparison Codebase Analysis

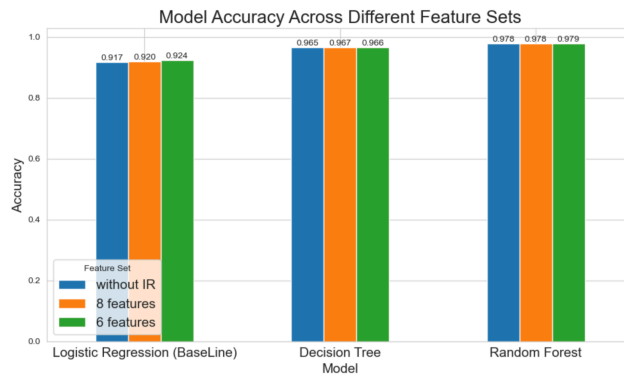


Image 4: Bar plot for Model Accuracy Across Different Feature Sets based on Supplementary & Extended Analysis

5.3. Plots and Visualisation

There are several data visualisation techniques used for data analysis in this project. A Bar chart and pie chart were used to show the distribution of Stellar objects such as stars, galaxies and QSO and to compare the model accuracy across different feature sets and comparison between models. A boxplot was used to check the outliers. Histogram was used to examine feature distributions. Heatmap was used for checking multicollinearity exists among the features. A seaborn pairplot was used to check how three objects are distributed.

6. Analysis & Discussion (interpretation, error analysis, limitations)

6.1. Analysis on performance comparison between models and baseline (Logistic Regression)

After evaluating the metrics across various models, XGBoost model has the best performance with an accuracy of 97.33%, precision of 97.29%, recall of 97.33% and F1_score of 97.30%. So, this model shows the effectiveness at capturing the complex relationships in the dataset.

Gradient Boosting, Random Forest and Decision Tree models have the better performance. In Gradient Boosting, 97.37% of accuracy and in Random Forest and Decision Tree, 95.87% respectively. However, these models show high accuracy, they do not reach the performance of XGBoost.

Logistic Regression (Baseline Model) has good performance with an accuracy of 91%, it provides a strong benchmark. However, when compared to XGBoost, it is not a good idea to use for complex datasets.

K-Nearest Neighbours (KNN) (model has a bad performance with an accuracy of 89.78%. This result specifies that KNN underperforms with large and complex datasets like SDSS and it may be captured well through.

The AdaBoost model has poor performance and 91.53% and it makes the least effective model for this dataset. This model could not capture the underlying patterns as effectively as the tree-based models.

6.2. Confusion Matrix

Our project uses various models to compare the performance and confusion matrix. After comparing the models, the confusion matrix of XGBoost, and Gradient Boosting, their classification performance in Star and Quasar (QSO) is the strongest.

6.3. Interpretation

XGBoost's capacity to capture complicated, non-linear correlations in the data is demonstrated by its persistent outperformance of similar models across all requirements.

Decision Tree and Random Forest models performed well and provide the same results, suggesting that ensemble approaches are more reliable than single decision trees.

KNN and AdaBoost cannot perform very well and it manages the dataset's feature complexity and class imbalance.

6.4. Error Analysis

When first performing data preprocessing, outliers were handled before applying normalization. The resulting data was used for model training.

Through observation, first removed a data point with extreme outliers (where all three feature values u, g, and z were -9999). Next, the IQR method was used to eliminate outliers from the feature columns. Finally, perform normalization processing.

The results show that the logistic regression achieved an accuracy of 0.59 and the confusion matrix showed that all test samples were classified as GALAXY, while the decision tree achieved an accuracy of 0.43.

The poor performance of this model is not due to the model itself, it is due to the data. This is because the preprocessing pipeline excessively removed outliers and then performed normalization on the resulting data.

This leads to a significant distortion in the data distribution and reduced inter-class differences, making it difficult for both linear and tree-based models to effectively distinguish between the three types of celestial bodies.

6.5. Limitations

The current outlier handling approach does not filter and delete data based on the number of outliers in a single record; instead, it deletes them directly.

There is no explanation for why the model's performance decreased after removing extreme outliers (where all three feature values u, g, and z were -9999).

7. Ethical & Legal Considerations

7.1. Data Protection and Privacy

The Stellar Classification Dataset – SDSS17 is available in public and it does not include any personal data. The dataset contains only astronomical observation including objects and features such as alpha, delta, u, g, r, I, z, redshift. The identifiers like obj_ID, field_ID, plate, etc. are referred to telescope observation and not related to personal information.

7.2. Bias and Fairness

In this case, bias and fairness concern human characteristics as gender or race, but instead to the equitable treatment of each object class by the models: galaxies, stars, and quasars. The SDSS dataset is slightly imbalanced, as certain classes are considerably more common than others. This can lead to models favoring the majority class and underperforming minority classes, such as quasars. This can be reflected in the confusion metrics and class metrics, where stars and galaxies tend to be more correctly classified than quasars. To reduce this, weighted precision, recall and F1 score metrics were used in the evaluation, rather than overall accuracy, to ensure that all classes were included in the quality metric. However, there is some bias, as quasars tend to be misclassified more often, particularly by less robust models like kNN and AdaBoost.

7.3. Responsible Use

Models which are used in this project are to help astronomers by collaboration with Data Scientists to identify and classify the space objects such as Star, Galaxy, and Quasar automatically using the real telescope data. This model can help astronomers to handle numerous volumes of data, thus they can focus more and use their efforts on comprehensive investigation. However, human verification is still essential, especially for edge cases and rare objects.

8. Conclusion & Future Work

8.1. Conclusion

As a conclusion, this Stellar Classification project shows how various machine learning can make the exploration of space faster, smarter and more efficient in astronomy research. We trained various kinds of models to identify the space objects easily. It can assist astronomers for their space exploration to handle huge volumes of data and more focus on their research, and it shows how data science and astronomy can collaborate.

8.2. Future Work

For further improvement, the project can extend in various directions to improve performance and effectiveness in real world research areas. We can explore more on deep learning models as neural networks, to handle large-scale astronomy datasets. At the same time, we can explore more on image processing approaches and combine them with the current features in models. It will further investigate the impact of extreme outliers on results and explore more granular handling of outliers.

9. Bibliography

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